

Structure Theorem for finitely generated R -modules, R PID

Uniqueness To show: if

$$\begin{aligned} M &\simeq R/I_1 \oplus \dots \oplus R/I_k \oplus E \\ M &\simeq R/I'_1 \oplus \dots \oplus R/I'_k \oplus E' \end{aligned}$$

then $k = k'$, $\forall i, I_i \simeq I'_i, E \simeq E'$.

Proof $M_{\text{tors}} \subset M, M_{\text{tf}} = M/M_{\text{tors}}$. Their formation is canonical: If $M \simeq M'$ then $M_{\text{tors}} \simeq M'_{\text{tors}}$ ($\bigoplus_i R/I_i \simeq \bigoplus_j R/I'_j$) and $M_{\text{tf}} \simeq M'_{\text{tf}}$ ($E \simeq E'$).

Reduce to the case $E = \{0\} = E'$. $M \simeq M' \implies M[p^\infty] \simeq M'[p^\infty]$ for $\forall p$. Pass from invariant factors to elementary factors $\implies$ it suffices to show: given a prime p in R , if

$$M := \bigoplus_{i=1}^k R/(p^{b_i}) \simeq \bigoplus_{i=1}^{k'} R/(p^{b'_i}) =: M'$$

then $k = k'$, and $\forall i, b_i = b'_i$.

Note that

$$M[p] = \bigoplus_i (R/(p^{b_i}))[p] = \bigoplus_i p^{b_i-1} R/p^{b_i} R \simeq \bigoplus_i R/(p)$$

is naturally $R/(p)$ -vector space (Recall: $R/(p)$ is a field), $\dim_{R/(p)} M[p] = k$.

More generally: $\forall j \geq 0$,

$$p^j M \simeq \bigoplus_i (\text{image of } p^j R \text{ in } R/p^{b_i} R) = \bigoplus_{i:j < b_i} p^j R/p^{b_i} R \simeq \bigoplus_{i:j < b_i} R/p^{b_i-j} R$$

$\implies (p^j M)[p]$ has $\dim_{R/(p)} =$ the number of i s.t. $j < b_i =: \nu_j$. Likewise get ν'_j for M' .

$M \simeq M' \implies \forall j, (p^j M)[p] \simeq (p^j M')[p]$, i.e. $\nu_j = \nu'_j$. Magic gives $\forall i, b_i = b'_i$. □

Remains to prove the Lemma: R PID, E free module with rank $= n \in \mathbb{Z}_{\geq 0}, N \subset E$ submodule $\implies \exists$ basis $f_1, \dots, f_n$ of $E, d_1 \mid \dots \mid d_n$ in R s.t. $\{d_i f_i \mid d_i \neq 0\}$ is a basis of N .

§ Smith normal form of matrices

R commutative ring, $n, m \in \mathbb{Z}_{\geq 1}, M$ R -module.

For $e_1, \dots, e_n \in M, x_1, \dots, x_m \in M, A = (a_{ij})_{i,j} \in M_{n \times m}(R), (x_1 \dots x_m) = (e_1 \dots e_n)A$ means:

$$x_j = \sum_{i=1}^n e_i a_{ij}, \quad \forall j = 1, 2, \dots, m.$$

Facts

- $(e_1 \dots e_n)(AB) = ((e_1 \dots e_n)A)B$,
- $(e_1 \dots e_n)1_{n \times n} = (e_1 \dots e_n)$,
- If $e_1, \dots, e_n \in M$ are linearly independent, then the A s.t. $(x_1 \dots x_n) = (e_1 \dots e_n)A$ is uniquely determined by x_i and e_j .

$$\text{GL}(n, R) := \{A \in M_{n \times n}(R) \mid \exists B \in M_{n \times n}(R), AB = 1_{n \times n} = BA\} = \{A \in M_{n \times n} : \det A \in R^\times\}.$$

Facts

- If $(y'_1 \dots y'_n) = (y_1 \dots y_n)P$, $P \in \text{GL}(n, R)$, then $\sum Ry_i = \sum Ry'_i$.
 - $\supset$ is clear. Multiply P^{-1} on both sides to get $\subset$.
- Assume M free, $e_1, \dots, e_n$ basis of M . $(e'_1 \dots e'_n) = (e_1 \dots e_n)A$, A unique. Then $e'_1, \dots, e'_n$ form a basis of $M \iff A \in \text{GL}(n, R)$.
 - $\implies$: $\exists! B$ s.t. $(e_1, \dots, e_n) = (e'_1, \dots, e'_n)B \implies AB = 1$. Similarly $BA = 1$.
 - $\impliedby$: $\sum Re'_i = \sum Re_i = M$. Linear independence: if $\sum e'_i b_i = 0$ i.e.

$$(e'_1 \mid \dots \mid e'_n) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = (e_1 \mid \dots \mid e_n) A \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0,$$

hence $A^t(b_1 \mid \dots \mid b_n) \implies \forall i, b_i = 0$. □

Let E free with basis $e_1, \dots, e_n$, $N \subset E$ submodule generated by $x_1, \dots, x_m$. Take the unique $A \in M_{n \times m}(R)$ s.t.

$$(x_1 \mid \dots \mid x_m) = (e_1 \mid \dots \mid e_n)A$$

Observe: $A \rightarrow AP$, $P \in \text{GL}(n, R) \iff$ replace $(x_1 \mid \dots \mid x_m)$ by $(x'_1 \mid \dots \mid x'_m) = (x_1 \mid \dots \mid x_m)P \iff$ change of generators of N . Similarly $A \rightarrow QA$ represents a change of basis of E .

Theorem (H. J. S. Smith) Let R PID, $A \in M_{n \times m}(R)$, then $\exists P \in \text{GL}(m, R)$, $Q \in \text{GL}(n, R)$ s.t.

$$A = Q \text{diag}(d_1, d_2, \dots)P$$

where $d_1 \mid d_2 \mid \dots$ in R .

Proof of Lemma Given E free, say with a basis $e_1, \dots, e_n$, $N \subset E$ submodule.

- N is finitely generated (omitted).
- Take generators $x_1, \dots, x_m$ of N . Take A s.t. $(x_1 \mid \dots \mid x_m) = (e_1 \mid \dots \mid e_n)A$, $A = Q \text{diag}(d_i)P$.
Hence $\underbrace{(x_1 \mid \dots \mid x_m)P^{-1}}_{\text{generators of } N} = \underbrace{(e_1 \mid \dots \mid e_n)Q \text{diag}(d_i)}_{\text{basis of } E}$
 $\implies N$ is generated by $x'_i = d_i f_i, d_1 \mid d_2 \mid \dots$ □

Proof of Smith's Theorem Assume $R = \mathbb{Z}$ or $F[X]$, F field. Start with the case of $\mathbb{Z}$.

Among all matrix entries of all QAP ($Q \in M_{n \times n}(R)$, $P \in M_{m \times m}(R)$), take the one with the minimal $|\cdot| \neq 0$. Can move it to $(1, 1)$ by multiplying permutation matrices, so may assume it is a_{11} .

Claim: $a_{11} \mid a_{1j}, \forall j > 1$; $a_{11} \mid a_{i1}, \forall i > 1$.

- If not, say $a_{1j} = a_{11}d + r, 0 < r < |a_{11}|$, can use elementary column operation $a'_{1j} \leftarrow a_{1j} - da_{11}$ to get r at $(1, j) \implies$ contradiction!

$\implies$ may assume $a_{1j} = a_{i1} = 0, \forall i, j \neq 1$. Define $d_1 := a_{11}$.

Induction: may multiply A by $\text{diag}(1, A')$ where $A' \in M_{(n-1) \times (n-1)}(R)$, so may assume $A = \text{diag}(d_i)$ where $d_2 \mid d_3 \mid \dots$. Remains to show $d_1 \mid d_2$.

- If not, $d_2 = d_1q + r, 0 < r < |d_1|$, enforce the following:
 - column 2 minus q times of column 1,
 - row 2 adds row 1.

Now $(2, 2)$ -entry is r , contradicting the choice of $d_1 = a_{11}$.

For $R = F[X]$: consider the entry with minimal degree. □

General PID R : See Exercise.

Definition R commutative ring. We say R is **Noetherian** if $\forall$ chain of ideals $I_1 \subset I_2 \subset \dots$ stabilizes, i.e. $\exists i_0$ s.t. $i \geq i_0 \implies I_i = I_{i_0}$.

We say an R -module M is Noetherian if $\forall$ chain of submodules $M_1 \subset M_2 \subset \dots$ stabilizes.

R is a Noetherian ring $\iff R$ is Noetherian as an R -module.

- submodules of R = ideals.

Recall PID are Noetherian.

Observe that Noetherian module M are finitely generated,

- otherwise $Rx_1 \subsetneq Rx_1 + Rx_2 \subsetneq Rx_1 + Rx_2 + Rx_3 \subsetneq \dots$

Proposition Assume R commutative ring, M R -module, $M' \subset M$ submodule, $M'' := M/M'$. Then M is Noetherian $\iff M', M''$ are both Noetherian.

Corollary If M_1, M_2 are Noetherian, then so is $M_1 \oplus M_2$.

- $\frac{M_1 \oplus M_2}{M_1} \simeq M_2$.

Corollary R PID, E free R -module, $\text{rank} = n \in \mathbb{Z}_{\geq 0} \implies E$ is Noetherian (so is its submodule, hence all its submodules are finitely generated).

- $E \simeq R \oplus \dots \oplus R$, R Noetherian.

Proof of Proposition

- $\implies$ Easy.
- $\impliedby$: Given $M_1 \subset M_2 \subset \dots$ in M , set

$$M'_i := M_i \cap M', M''_i := (\text{image of } M_i \text{ in } M'') = (M_i + M')/M'$$

$\implies \exists i_0, \forall i \geq i_0, M'_i = M'_{i+1}, M''_i = M''_{i+1}$. To show: $i \geq i_0, M_i \supset M_{i+1}$.

$$\forall x \in M_{i+1}, \exists y \in M_i \text{ s.t. } \underbrace{x + M'}_{\in M'_{i+1}} = \underbrace{y + M'}_{\in M'_i} \implies x - y \in M' \cap M_{i+1} = M'_{i+1} = M'_i \subset M_i,$$

hence $x \in M_i$. So $M_{i+1} \subset M_i$. □